

Kodbud Python Internship

Code Submission — Part 1 of 2

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Task 1 — Simple Calculator (CLI)

A command-line calculator that handles addition, subtraction, multiplication, and division. Uses input() for user interaction and if-elif-else for operation routing. Includes division-by-zero protection.

■ Source Code

```
# Task 1: Simple Calculator (CLI)
# Concepts: input(), float conversion, if-elif-else, arithmetic operators

def calculator():
    print("=" * 40)
    print(" Simple Calculator")
    print("=" * 40)
    print("Operations: + - * /")
    print()
    try:
        num1 = float(input("Enter first number : "))
        op = input("Enter operator (+, -, *, /): ").strip()
        num2 = float(input("Enter second number : "))
    except ValueError:
        print("Invalid input! Please enter numeric values.")
        return

    if op == "+":
        result, symbol = num1 + num2, "+"
    elif op == "-":
        result, symbol = num1 - num2, "-"
    elif op == "*":
        result, symbol = num1 * num2, "*"
    elif op == "/":
        if num2 == 0:
            print("Error: Division by zero is not allowed!")
            return
        result, symbol = num1 / num2, "/"
    else:
        print(f"Unknown operator '{op}'. Please use +, -, *, or /.")
        return

    print()
    print(f" {num1} {symbol} {num2} = {result}")
    print("=" * 40)

# Entry point
while True:
    calculator()
    again = input("\nCalculate again? (yes/no): ").strip().lower()
```

```
if again not in ("yes", "y"):
    print("Goodbye!")
    break
```

■ Sample Run

```
Enter first number : 15
Enter operator : /
Enter second number : 4
```

```
15.0 / 4.0 = 3.75
```

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```

Task 2 — Number Guessing Game

Uses Python's random module to pick a secret number between 1 and 100. The player receives 'Too high' / 'Too low' hints on each guess. Attempt count is tracked and displayed on success. Limited to 10 tries.

■ Source Code

```
# Task 2: Number Guessing Game
# Concepts: random module, while loop, int conversion, attempt counter

import random

def guessing_game():
    print("=" * 45)
    print(" Number Guessing Game")
    print("=" * 45)
    print("I've picked a secret number between 1 and 100.")
    print("Try to guess it!\n")

    secret = random.randint(1, 100)
    attempts = 0
    max_tries = 10

    while attempts < max_tries:
        try:
            guess = int(input(f"Attempt {attempts+1}/{max_tries} - Your guess: "))
        except ValueError:
            print(" Please enter a whole number.\n")
            continue

        attempts += 1

        if guess < 1 or guess > 100:
            print(" Out of range! Guess between 1 and 100.\n")
            attempts -= 1
            continue

        if guess == secret:
            print(f"\n Correct! The number was {secret}.")
            print(f" You guessed it in {attempts} attempt(s).")
            if attempts == 1:
                print(" Incredible – first try!")
            elif attempts <= 3:
                print(" Impressive!")
            print("=" * 45)
            return
        elif guess < secret:
            diff = secret - guess
            hint = "Very close! " if diff <= 5 else ""
            print(f" Too low! {hint}Try higher.\n")
        else:
            diff = guess - secret
            hint = "Very close! " if diff <= 5 else ""
            print(f" Too high! {hint}Try lower.\n")
```

```
print(f"\n Game over! The secret number was {secret}.")
print(" Better luck next time!")
print("=" * 45)

# Entry point
while True:
    guessing_game()
    play_again = input("\nPlay again? (yes/no): ").strip().lower()
    if play_again not in ("yes", "y"):
        print("Thanks for playing!")
        break
```

■ Sample Run

```
Attempt 1/10 – Your guess: 50
Too high! Try lower.

Attempt 2/10 – Your guess: 25
Too low! Try higher.

Attempt 3/10 – Your guess: 37
Correct! The number was 37.
You guessed it in 3 attempt(s).
Impressive!
```